

Avatar-Based Elderly Companion

Problem Statement

Current LLM companions for older adults primarily rely on short-term conversation history or similarity-based retrieval. They often forget important life experiences, fail to maintain a coherent sense of identity across months of interaction, and respond generically to emotional situations. This reduces trust, engagement, and long-term companionship.

Proposed Idea

Develop an avatar-based conversational companion powered by an Identity-Aware Narrative Memory framework. Instead of storing isolated facts, the system extracts life events, relationships, values, emotions, and roles from conversations and organizes them into a structured Life Story Graph. During each conversation, the system retrieves identity-relevant memories, ranks them using a reliability score, and injects them into the prompt of a frozen instruction-tuned LLM (e.g., Qwen 3 but we will start with llama 3.1 first).

Research Novelty and Authenticity

- Narrative memory instead of flat fact memory.
- Identity-aware retrieval rather than semantic similarity alone.
- Reliability-guided memory ranking to reduce contradictory or low-confidence memories.
- Long-term life-story continuity for avatar companions.
- New evaluation metrics such as Identity Consistency, Narrative Recall, and Story Continuity.

Expected Contributions

- Novel Identity-Aware Narrative Memory framework.
- Life Story Graph representation for elderly conversations.
- Reliability-guided retrieval algorithm.
- Avatar architecture for long-term companionship.
- Benchmark comparison against baseline LLM and standard RAG.
- Improved Emotional support: Provides more meaningful and empathetic interactions by recalling relevant memories during conversations, helping reduce feelings of loneliness and social isolation among older adults.

System Architecture

User

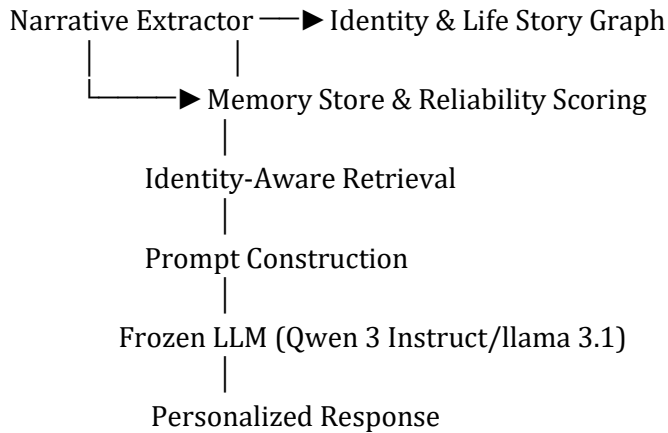


Avatar Interface



Backend API





Pipeline

1. User interacts with the avatar.
2. Conversation is transcribed and stored.
3. Life events, relationships, values, emotions, and temporal information are extracted.
4. Extracted information updates the Life Story Graph.
5. Relevant memories are ranked by relevance, identity alignment, recency, and reliability.
6. Top memories are inserted into the LLM prompt.
7. The LLM generates an empathetic, identity-consistent response.
8. Conversation is evaluated and important new memories are stored.

Implementation Plan

Base LLM: Llama 3.1 Qwen 3 Instruct

Memory DB: ChromaDB/FAISS

Backend: FastAPI

Frontend: React + Avatar

Datasets: LoCoMo, PersonaChat, Empathetic Dialogues, DailyDialog, Healthy Aging tips.

NOTE: After our prototype is working, we rerun the same experiments with Qwen and compare it with our framework: ---for the SCIE paper journal.